

solar cells, which have an efficiency of about 20 to 25 per cent. Their higher price-to-performance ratio has limited their use in aerospace.

One can't merely stack different solar cells on top of each other because the different materials are structurally incompatible and so charges can't pass through them. To solve this problem, in current multi-junction solar cells, heavily doped metals are used to create a tunnel junction between the various layers—adding significant complexity, which increases the cost. The tunnel junction enables electrical connectivity by acting as a metal-to-metal connection.

There is another technique under development by North Carolina State University in the USA, in which the existing metal contacts of the individual cells are covered with indium films. The indium films bond to themselves easily at room temperature under low pressure.

This method, if commercialised, would benefit the manufacturers who make multiple types of single-junction solar cells using say silicon, III-V materials, etc. Single junction solar cells made of two different materials can be mechanically stacked and electrically connected to produce MJ solar cells, as shown in Fig. 3.

This technique enables the manufacturers to use the existing plant and equipment and simply tweak their existing products slightly to produce more efficient multi-junction solar cells rather than having to create new products, resulting in considerable cost reduction.

How a multi-junction cell operates?

The behaviour of MJ solar cells is inherently different than that of conventional single-junction (SJ) solar cells. The internal series connection of several junctions with different band gaps makes these devices more sensitive to incident spectral irradiance. The energy needed to excite the electron across the junction or gap is the band gap energy (in eV).

Only certain frequencies of light that have energy equal to or higher than eV of a particular material can excite the electrons across the junction. However, the light that has lower

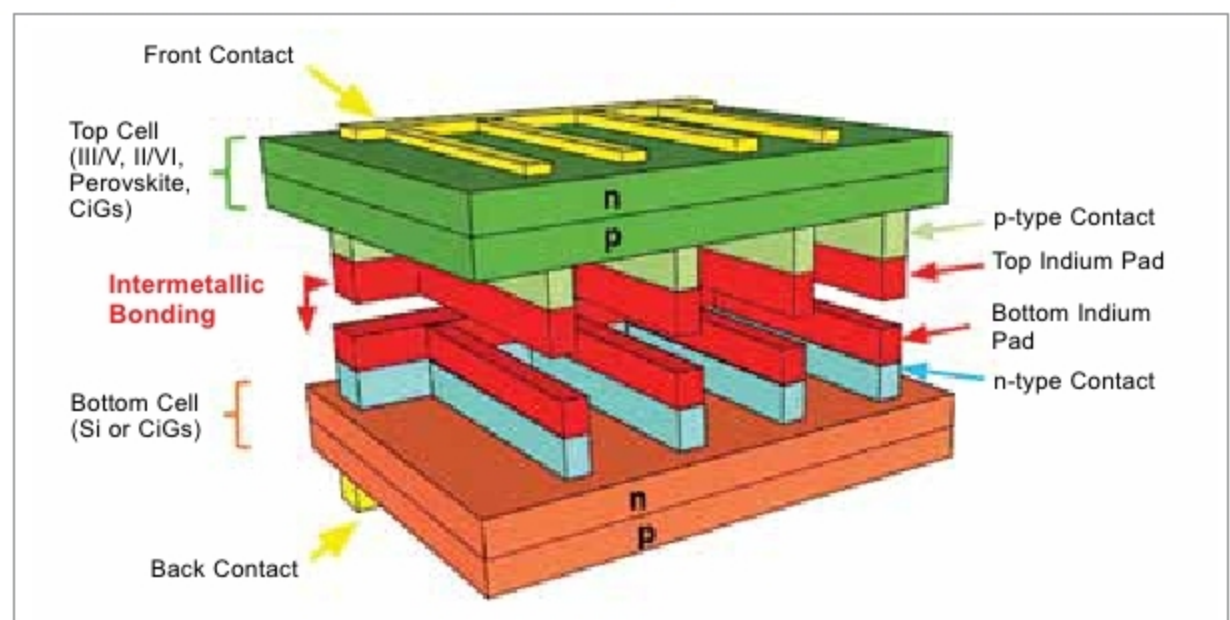


Fig. 3: New technique of manufacturing MJ solar cells

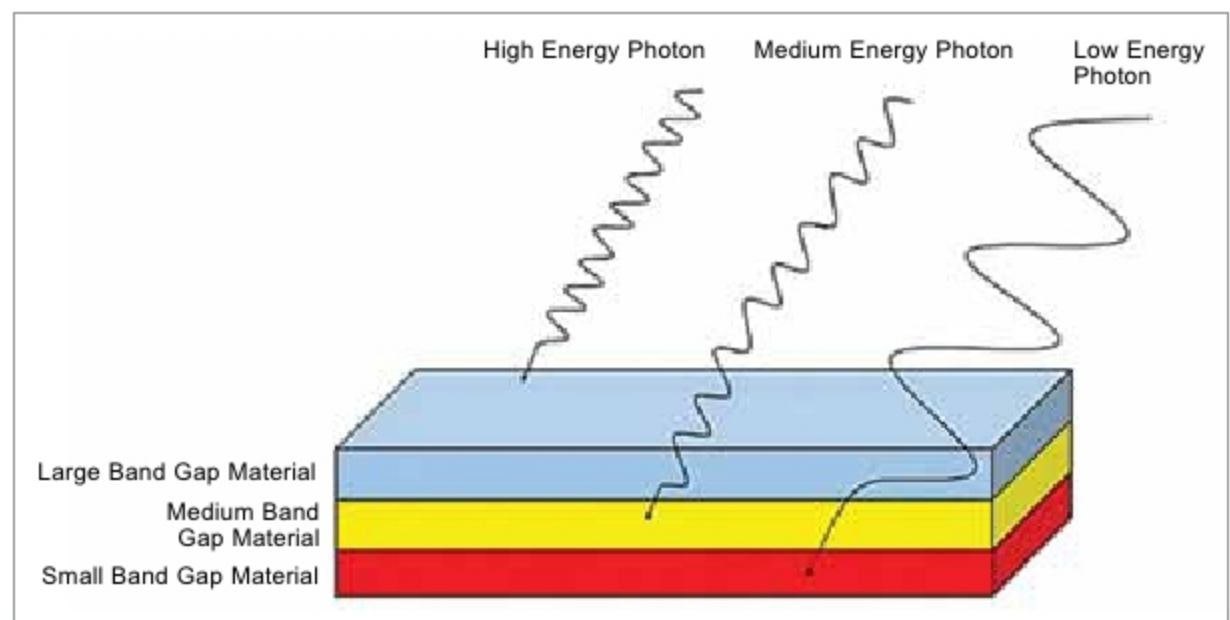


Fig. 4: Mechanism of absorption of light having different spectrum

energy will pass through.

The size of the band gap dictates the energy generated by exciting an electron. A smaller band gap will result in lower energy. Therefore there is a balance between having a small band gap that will utilise many photons and having a large band gap that will utilise fewer photons and yet generate optimum energy. The optimum band gap is 1.4eV, and a single junction solar cell with this eV has a theoretical efficiency limit of 31 per cent approximately.

The actual efficiency can be greatly improved with the addition of multiple p-n junctions with multiple band gaps and working in tandem. With the various band gaps, the higher energy photons can be better utilised by materials with large band gaps, while the lower energy photons can be utilised by those with smaller band gaps. This is the basic principle of operation of a multi-junction cell. A multi-junction cell layers the materials in descending

order, with the largest band gaps on top and the smallest on bottom, which creates a 'photon sorting' effect.

Photons with high energy are absorbed by the top layer more efficiently than if they were absorbed first by the bottom layer. Other spectrums of light having middle energy photons are transmitted through the top layer and absorbed by the middle layer photovoltaics, while the lower energy photons continue to pass through and are ultimately captured by the lowest layer of the solar cell. By deploying multiple junctions, the light that would otherwise have been wasted as heat can now be converted into electricity (refer Fig. 4).

Materials used in multi-junction solar cells

The multi-junction cells leverage the properties of semiconductors comprising elements in the III and V columns of the periodic table. The most common material used is gallium